FPGA Parallelism – Lab 4

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HLS synthesized latency:

```
================================================================
== Performance Estimates
================================================================
+ Timing:
    * Summary:
    +--------+----------+---------+-----------+
    | Clock  |  Target  | Estimated| Uncertainty|
    +--------+----------+---------+-----------+
    |ap_clk  | 10.00 ns|  7.300 ns|     2.70 ns|
    +--------+----------+---------+-----------+

+ Latency:
    * Summary:
    +---------+---------+----------+----------+--------+--------+----------+
    |  Latency (cycles) |  Latency (absolute) |     Interval    | Pipeline |
    |   min   |   max   |    min   |    max   |  min   |   max  |   Type   |
    +---------+---------+----------+----------+--------+--------+----------+
    |   133656|   133656|  1.337 ms|  1.337 ms|  133634|  133634|  dataflow|
    +---------+---------+----------+----------+--------+--------+----------+
```

Co-simulation latency and clock period:

```
+----------+---------+-----------------------------------------+-----------------------------------------+----------------------+
|          |         |            Latency(Clock Cycles)         |            Interval(Clock Cycles)        | Total Execution Time |
+   RTL    +  Status +-----------------------------------------+-----------------------------------------+    (Clock Cycles)    +
|          |         |      min     |      avg     |      max     |      min     |      avg     |      max     |                      |
+----------+---------+-----------------------------------------+-----------------------------------------+----------------------+
|      VHDL|       NA|            NA|            NA|            NA|            NA|            NA|            NA|                    NA|
|   Verilog|     Pass|        132981|        132981|        132981|            NA|            NA|            NA|                132981|
+----------+---------+-----------------------------------------+-----------------------------------------+----------------------+
```

Post-implementation resource utilization:

```
#=== Post-Implementation Resource usage ===
SLICE:            0
LUT:           7474
FF:            5556
DSP:             26
BRAM:             4
URAM:             0
LATCH:            0
SRL:            177
CLB:           1604

#=== Final timing ===
CP required:                     10.000
CP achieved post-synthesis:      5.693
CP achieved post-implementation: 7.230
Timing met
```

Final speedup ratio calculation:

132981 cycles * 7.23 post-imp. clock period = 0.96145 ms

51.962 / 0.96145 = <u>54.0453 speedup</u>